

Venkata Sai Rahul Unnam

Python Software Engineer | Agentic AI Systems

Denton, TX • Email: uvsvrahul1234@gmail.com • Phone: (940) 629-8856 • LinkedIn: [linkedin.com/in/rahul-unnam](https://www.linkedin.com/in/rahul-unnam)
GitHub: github.com/uvsvrahul1234 • LeetCode: leetcode.com/u/UvsRahul • Portfolio: <https://uvsvrahul1234.github.io/Portfolio/>

Profile Summary

Results-driven Software Engineer with hands-on experience in designing, building, and improving production-grade Agentic AI systems and scalable MLOps pipelines. Proven track record in agent orchestration, latency optimization, and deploying robust, production-ready Python applications on scalable cloud infrastructure. Expertise in Natural Language Understanding (NLU), LLM self-reflection, and sandboxed code execution. Skilled in AWS, Apache Airflow, LangChain, LSTM architectures, and DevSecOps practices with strong cross-functional collaboration experience.

Technical Skills

Programming: Python (Expert), SQL (Advanced: Window Functions, CTEs), Java, Bash, RESTful API Design, FastAPI
Data Engineering: ETL/ELT Pipelines, AWS Glue, Apache Airflow (MWAA), dbt, Spark, Kafka, AWS Lambda, Snowflake, PostgreSQL
Analytics & Observability: Elasticsearch, Kibana, Logstash (ELK Stack), Advanced SQL, Tableau, Power BI, Grafana, MLflow
Data Modelling: Dimensional Modelling (Star/Snowflake Schema), Entity-Relationship (ER) Diagrams, Data Warehousing
Cloud: AWS (EC2, IAM, ECS, S3), EventBridge, Redshift, Glue, Secrets Manager, Docker, Kubernetes, Terraform, Snowflake, GitHub Actions, Code Reviews, Azure, Streamlit, Apache Airflow
AI/ML: LangChain, LangGraph, RAG Architectures, ChromaDB, Prompt Engineering, OpenAI API Integrations, PyTorch, TensorFlow, Scikit-Learn, Time-Series Forecasting, HuggingFace, Agent Orchestration, LLM Self-Reflection, Natural Language Understanding

Work Experience

Graduate Research Assistant | Software Engineer | Environmental Lab May 2025 – May 2026
University of North Texas | Denton, TX

- Authored a first-author IEEE research paper on Physics-Informed Neural Network (PINN) by embedding partial differential equations (PDEs) directly into the custom deep learning loss function, mitigating gradient explosion and increasing long-term forecast stability while maintaining a high standard of operational excellence.
- Developed a robust spatial-temporal data pipeline to process multi-year and high-dimensional environmental grid data. Conducted extensive feature engineering, normalization, and dimensionality reduction on unstructured spatial-grid datasets for sequence modeling.
- Evaluated and benchmarked deep learning architectures (LSTM, PINN) against classical statistical baselines (SARIMAX) and optimized latency and hyperparameters to achieve an 18% reduction in Root Mean Squared Error (RMSE), significantly outperforming traditional time-series models in forecasting accuracy.

Senior Systems Engineer - Onsite Dec 2020 – Jul 2024
Infosys Ltd. | Delhi, India

- Architected serverless ETL/ELT workflows using AWS Lambda and Airflow to ingest and orchestrate multi-source enterprise data streams, resulting in reduced data latency by 40% and enabling real-time KPI tracking for executive-level Power BI dashboards.
- Designed and deployed production-grade end-to-end predictive modeling workflow using XGBoost and LSTM architectures for industrial underflow density forecasting and integrated Optuna, GridSearchCV for automated hyperparameter tuning to achieve 1.8% MAPE (vs Random Forest 3.4%, ANN 7.3%) while reducing training time by 40%.
- Spearheaded the migration of legacy on-premises data to Amazon Redshift, implementing dimensional modeling (Star Schema) that accelerated query execution time by ~3x and empowered the BI team to run complex analytics reports without timeouts.
- Engineered a scalable NLU pipeline integrating OpenAI API to parse high-volume (30GB+) unstructured log data into structured enterprise knowledge graphs for trend analysis, guiding client resource allocation decisions and eliminating 6 hours of manual review per week.
- Designed custom search and observability configurations using the ELK Stack (Elasticsearch, Logstash, Kibana) to parse real-time network telemetry, uncovering operational bottlenecks and resulting in a 30% reduction in software release cycle times.
- Maintained strict attention to detail by engineering automated data quality anomaly checks (Python) within CI/CD pipelines, reducing downstream BI reporting errors by 24% and ensuring strict data governance for client regulatory compliance audits.

Projects

State-Aware Enterprise Graph Agent & Semantic Caching Layer | [Project Link](#)

- Architected a high-performance Python backend (FastAPI) to orchestrate an AI agent capable of translating natural language into complex Cypher queries, autonomously navigating deeply interconnected enterprise knowledge graphs.

- Engineered a distributed semantic caching layer using Redis and vector embeddings to intercept redundant conversational queries, reducing query latency by 98% (from ~2.5s to <15ms) and cutting LLM token consumption by 40% under simulated workloads.
- Implemented automated conversational memory pruning and state management within a deterministic LangGraph state machine, preventing context-window overflow during multi-turn data retrieval tasks.
- Built robust validation loops to catch, parse, and self-correct AI-generated database syntax errors dynamically before execution against the graph database, ensuring high reliability in production environments.
- Developed comprehensive benchmarking scripts and containerized the execution environment using Docker Compose to validate system performance and latency optimizations under simulated enterprise traffic loads.

ARM (Autonomous Reliable Mechanism) - AI Software Engineering Agent | [Project Link](#)

- Architected a deterministic state machine using LangGraph and Python to handle complex agent orchestration, autonomously resolving multi-file enterprise repository bugs.
- Engineered a sandboxed code execution environment using a "Self-Healing" Docker loop to test AI-generated patches dynamically against native test suites.
- Developed LLM self-reflection and improvement capabilities by feeding traceback compiler errors back to the model for iterative correction, achieving a 65% autonomous bug resolution rate on SWE-bench Lite.
- Validated model logic against human ground-truth data using SHAP feature attribution and AST (Abstract Syntax Tree) edit distance, proving the agent's structural reliance on execution feedback.

Education

Master of Science in Data Science (Summa Cum Laude), GPA 4.0/4.0

Aug 2024 – May 2026

University of North Texas | Denton, TX

- Publication: Benchmarking Scientific Machine Learning Models for Air Quality Data (2026) | [Article](#)

Bachelor of Technology in Mechanical Engineering, GPA 3.4/4.0

Aug 2016 – May 2020

Vellore Institute of Technology | Vellore, TN

- Chief Mechanical Engineer of Pravega Racing Formula SAE team.
- Treasurer of Creativity Club.

Certifications

Advanced Certification in Data Science & AI - IIT Madras | [Link](#)

DevOps Foundation - DevOps Institute • **Infosys Certified DevSecOps Professional** | [Link](#)